Inference with Unknown Variability

Lectures 22–23

QBA201 – Quantitative Business Analysis

Shi Qiu – Fall 2026

What changes when the population SD is unknown?

For independent observations from a normal population,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

This remains true whether or not we know σ.

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \quad \longrightarrow \quad T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

The studentized statistic follows t; the sample mean does not become t-distributed.

Read a t distribution

- Symmetric about zero, with heavier tails than the standard normal.
- Degrees of freedom: $\nu = n - 1$ for a one-sample mean.
- Smaller ν means more uncertainty from estimating σ.
- As ν increases, the t curve approaches the standard normal curve.

For the same tail probability, a t critical value exceeds the corresponding z value.

Read a t critical value

Define $t_{\alpha,\nu}$ by $\mathrm{P}(T > t_{\alpha,\nu}) = \alpha$.

Degrees of freedom	Upper tail 0.05	Upper tail 0.025
5	2.015	2.571
10	1.812	2.228
24	1.711	2.064
99	1.660	1.984
Normal limit	1.645	1.960

A 95% two-sided procedure needs 2.5% in each tail.

The one-sample t procedure

Let μ_0 be the hypothesized mean. Calculate

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad \nu = n - 1.$$

- Observations should be randomly sampled and independent.
- A normal population gives the exact t result.
- With sufficiently large samples, approximation may be reasonable; severe skewness and outliers need care.

Unknown σ alone is not enough to justify a t test.

Let the claim determine the tail

Alternative	Meaning	Reject when
$H_1 : \mu < \mu_0$	Mean is lower	$t < -t_{\alpha,\nu}$
$H_1 : \mu > \mu_0$	Mean is higher	$t > t_{\alpha,\nu}$
$H_1 : \mu \neq \mu_0$	Mean differs	$\lvert t \rvert > t_{\alpha/2,\nu}$

For a one-sided composite null, calculate the reference distribution at its boundary $\mu = \mu_0$.

Choose the alternative before looking at the sample result.

Same statistic, different decisions

A sample of $n = 100$ gives $t = -1.70$, with $\nu = 99$.

Alternative	p-value	Decision at $\alpha = 0.05$
$\mu < \mu_0$	0.0461	Reject H_0
$\mu \neq \mu_0$	0.0923	Do not reject H_0

At $\alpha = 0.10$, the two-sided test also rejects.

Specify both the alternative and the significance level when reporting a decision.

A salary claim: translate every statement

Constructed example

A random sample of 25 salaries has mean \$188,000 and sample SD \$20,000. Assume the population is normal. Test whether the population mean differs from \$180,000.

$$n = 25, \quad \bar{x} = 188{,}000, \quad s = 20{,}000, \quad \mu_0 = 180{,}000.$$

$$H_0 : \mu = 180{,}000, \qquad H_1 : \mu \neq 180{,}000.$$

Sample SD means σ is unknown: use t with 24 degrees of freedom.

Calculate, decide, interpret

$$\widehat{\mathrm{SE}}(\bar{X}) = \frac{20{,}000}{\sqrt{25}} = 4{,}000, \qquad t = \frac{188{,}000 - 180{,}000}{4{,}000} = 2.$$

With $\nu = 24$,

$$p = 2\,\mathrm{P}(T_{24} \geq 2) = 0.0569.$$

At $\alpha = 0.05$, do not reject H_0. At $\alpha = 0.10$, reject H_0.

At 5%, the sample does not provide sufficient evidence of a different mean. This does not establish equality.

Estimate the same mean with an interval

When σ is unknown, a $(1-\alpha)$ confidence interval is

$$\bar{x} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}.$$

For the same salary example, a 95% interval is

$$188{,}000 \pm 2.0639(4{,}000) = 188{,}000 \pm 8{,}255.6,$$

or approximately **$179,744 to $196,256**.

The interval contains $180,000, agreeing with the two-sided 5% test.

Now change the parameter: population variance

The sample variance estimates the population variance:

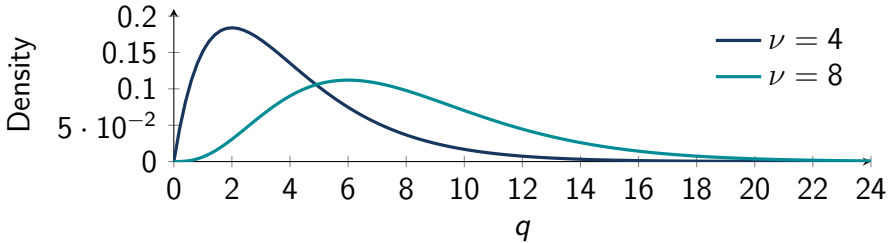
$$S^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}.$$

For independent observations from a normal population,

$$Q = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

This exact chi-square result requires a normal population. The statistic is not a z or t statistic.

The chi-square distribution is not symmetric



Its values are nonnegative; its shape depends on the degrees of freedom.

Do not use symmetric $\pm$ critical values for a variance test.

Choose the variance-test rejection region

Let q_p satisfy $\mathrm{P}(\chi^2_{n-1} \le q_p) = p$. Calculate $Q = (n-1)s^2/\sigma_0^2$.

Alternative	Reject H_0 when
$\sigma^2 > \sigma_0^2$	$Q > q_{1-\alpha}$
$\sigma^2 < \sigma_0^2$	$Q < q_{\alpha}$
$\sigma^2 \neq \sigma_0^2$	$Q < q_{\alpha/2}$ or $Q > q_{1-\alpha/2}$

For a two-sided 5% test, place 2.5% in each unequal-length tail.

Variance test: a complete example

Constructed example

A random sample of $n = 20$ normal-process measurements has $s^2 = 36$. Test $H_0 : \sigma^2 = 25$ against $H_1 : \sigma^2 \neq 25$ at 5%.

$$Q = \frac{19(36)}{25} = 27.36, \qquad \nu = 19.$$

The critical values are $q_{0.025} = 8.9065$ and $q_{0.975} = 32.8523$.

$$8.9065 < 27.36 < 32.8523.$$

Do not reject H_0: insufficient evidence that the variance differs from 25.

Compare variance with variance; keep squared units consistent.

Proportion inference uses a different standard error

For $H_0 : p = p_0$,

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}.$$

- Use p_0 in a test's null standard error.
- Check random sampling and independence.
- Check $np_0 \geq 5$ and $n(1 - p_0) \geq 5$ for the course approximation.

A proportion confidence interval instead estimates its SE using $\hat{p}$.

A compact inference checklist

1. Identify the parameter: μ, σ^2, or p.
2. Translate the claim into H_0 and H_1.
3. Check the sampling and distribution assumptions.
4. Calculate the statistic and its reference distribution.
5. Compare the correct p-value with α.
6. State a conclusion about the population, in context.

Do not reject is not the same as accept or prove.